

A TWO-PATH IMPROVEMENT FOR DIRECTED CONNECTIVITY BOUNDS

ABSTRACT. For a simple digraph on n vertices in which every local arc-connectivity, or every local vertex-connectivity, is at most $m - 1$, the bound $\lfloor n^2/4 \rfloor + 4(m - 1)(n - 1)$ was previously known. We observe that such a digraph avoids m directed paths of length two with common endpoints. A theorem of Huang and Lyu therefore improves the upper bound, for every fixed $m \geq 3$ and all sufficiently large n , to

$$\frac{n^2}{4} + (m - 1)n + O_m(1).$$

Thus the coefficient of the linear error term drops by a factor of four.

1. THE TWO-PATH THEOREM

All digraphs below are *strict*: they have no loops or parallel arcs, but a pair of opposite arcs is allowed. For distinct vertices u, v of a digraph D , write

$$p_2^D(u, v) := |\{x \in V(D) : ux, xv \in A(D)\}|.$$

Thus $p_2^D(u, v)$ is the number of directed u - v paths of length two. Let $P_{s,2}$ denote the union of s such paths with common initial and terminal vertices and distinct internal vertices.

For an integer $t \geq 2$, put

$$N(t) := \max \left\{ t^3 + 4t^2 + 3t + 4, \frac{17}{2}t^2 + 30t + 27 \right\},$$

$$g(n, t) := \left\lceil \frac{n+t}{2} \right\rceil \left\lfloor \frac{n-t}{2} \right\rfloor + tn + 1.$$

We use the following result of Huang and Lyu.

Theorem 1 (Huang–Lyu [1]). *Let $t \geq 2$ and $n \geq N(t)$. Then*

$$\text{ex}(n, P_{t+1,2}) \leq g(n, t).$$

More precisely, equality holds when $\lfloor (n-t)/2 \rfloor$ is odd, while for even $\lfloor (n-t)/2 \rfloor$ the value belongs to $\{g(n, t) - 1, g(n, t)\}$.

The form needed here is an immediate consequence.

Lemma 2 (Improved counting core). *Let $C \geq 2$ and $n \geq N(C)$. Suppose that a strict digraph D satisfies*

$$(1) \quad \mathbf{1}_{uv \in A(D)} + p_2^D(u, v) \leq C \quad \text{for all distinct } u, v \in V(D).$$

Then

$$(2) \quad |A(D)| \leq g(n, C).$$

In particular,

$$(3) \quad |A(D)| \leq \frac{n^2}{4} + Cn + 1 - \frac{C^2}{4} = \frac{n^2}{4} + Cn + O_C(1).$$

Proof. Condition (1) implies $p_2^D(u, v) \leq C$ for every ordered pair (u, v) . Hence D contains no copy of $P_{C+1,2}$, and Theorem 1 gives (2).

For the displayed estimate, observe that

$$\left\lceil \frac{n+C}{2} \right\rceil \left\lfloor \frac{n-C}{2} \right\rfloor \leq \frac{n^2 - C^2}{4}.$$

Substitution in the definition of $g(n, C)$ proves (3). □

2. APPLICATION TO BOUNDED LOCAL CONNECTIVITY

For a strict digraph D , let $\lambda_D(u, v)$ be the maximum number of arc-disjoint directed u - v paths, and let $\kappa_D(u, v)$ be the maximum number of internally vertex-disjoint directed u - v paths. Define $\ell_m^{\text{dir}}(n)$ and $k_m^{\text{dir}}(n)$ to be the largest number of arcs in an n -vertex strict digraph satisfying, respectively,

$$\lambda_D(u, v) \leq m - 1 \quad \text{or} \quad \kappa_D(u, v) \leq m - 1 \quad (u \neq v).$$

Theorem 3 (Sharper linear error term). *Let $m \geq 3$ and $n \geq N(m-1)$. Then*

$$(4) \quad \ell_m^{\text{dir}}(n) \leq k_m^{\text{dir}}(n) \leq \left\lceil \frac{n+m-1}{2} \right\rceil \left\lfloor \frac{n-m+1}{2} \right\rfloor + (m-1)n + 1.$$

Consequently,

$$(5) \quad \ell_m^{\text{dir}}(n), k_m^{\text{dir}}(n) \leq \frac{n^2}{4} + (m-1)n + O_m(1).$$

Proof. Fix distinct vertices u, v . The direct arc uv , when present, together with all directed paths $u \rightarrow x \rightarrow v$, is a pairwise arc-disjoint family: different middle vertices use different first and last arcs. It is also a pairwise internally vertex-disjoint family, since the two-step paths have distinct internal vertices and the direct path has none. Therefore either of the hypotheses

$$\lambda_D(u, v) \leq m-1, \quad \kappa_D(u, v) \leq m-1$$

implies

$$\mathbf{1}_{uv \in A(D)} + p_2^D(u, v) \leq m-1.$$

Lemma 2, with $C = m-1$, now gives the upper bound in (4). Finally, every arc-feasible digraph is vertex-feasible because $\kappa_D(u, v) \leq \lambda_D(u, v)$, so $\ell_m^{\text{dir}}(n) \leq k_m^{\text{dir}}(n)$. Equation (5) follows from (3). $\square$

The exact parity form of the new upper bound is occasionally convenient. With $t = m-1$,

$$g(n, t) = \frac{n^2}{4} + tn + 1 - \begin{cases} t^2/4, & n \equiv t \pmod{2}, \\ (t+1)^2/4, & n \not\equiv t \pmod{2}. \end{cases}$$

Corollary 4 (Comparison with the conjectured value). *For every fixed $m \geq 3$ and all sufficiently large n ,*

$$(6) \quad \left\lfloor \frac{(n+m-2)^2}{4} \right\rfloor \leq \ell_m^{\text{dir}}(n) \leq k_m^{\text{dir}}(n) \leq g(n, m-1).$$

Hence

$$\frac{n^2}{4} + \frac{m-2}{2}n + O_m(1) \leq \ell_m^{\text{dir}}(n) \leq k_m^{\text{dir}}(n) \leq \frac{n^2}{4} + (m-1)n + O_m(1).$$

Proof. The lower bound is the standard augmented bipartite construction: take all arcs from a part A to a part B , and give every vertex of B exactly $m-2$ cyclic in-neighbours inside B , with $|B| = \lceil (n+m-2)/2 \rceil$. It has $\lfloor (n+m-2)^2/4 \rfloor$ arcs and local arc-connectivity at most $m-1$. The remaining inequalities are Theorem 3. $\square$

Remark 5. Theorem 3 improves the coefficient $4(m-1)$ in the previously known bound

$$\left\lfloor \frac{n^2}{4} \right\rfloor + 4(m-1)(n-1)$$

to $m-1$, at the cost of requiring $n \geq N(m-1)$; compare [2, Theorems 1.11 and 1.12]. It does not prove the exact conjecture. The condition of being $P_{m,2}$ -free remembers only paths of length two, whereas bounded local connectivity also restricts the direct arc and all longer paths. That discarded information is precisely where a further improvement would have to enter.

REFERENCES

- [1] Z. Huang and Z. Lyu, *Extremal digraphs containing at most t paths of length 2 with the same endpoints*, Discrete Appl. Math. **388** (2026), 1–10.
- [2] L. Vandenbruwaene, *Maximising Edge Density Subject to Connectivity Constraints: Erdős Problem 915, from Proof to Search*, Master's thesis, KU Leuven, 2026.